Samuel DELÉGLISE

LPENS (équipe QUANTIC)
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Curriculum Vitae

Thèmes de recherche

Systèmes hybrides circuits quantiques–mécanique ; circuits supraconducteurs (fluxonium, haute impédance) ; métrologie et contrôle quantiques ; codes bosoniques, qubits de chat, stabilisation dissipative.

Positions et parcours

2026–	Chargé de recherche CNRS (CR1), LPENS (Paris), équipe QUANTIC – circuits hybrides supraconducteur–mécanique ; codes bosoniques.
2016–2025	Chargé de recherche CNRS (CR1), LKB (Paris), équipe OMQ – optomécanique quantique ; circuits hybrides supraconducteur–mécanique.
2012–2016	Chargé de recherche CNRS (CR2), LKB (Paris), équipe OMQ – optomécanique quantique.
2009–2012	Postdoctorant (bourse Marie Curie), EPFL (Lausanne) – optomécanique en micro-résonateurs. Dir. : T. Kippenberg.

Formation

2006–2009	Doctorat de physique quantique, Université Pierre et Marie Curie / ENS Paris. *« Preparation and reconstruction of non-classical field states in cavity quantum electrodynamics »* Laboratoire Kastler Brossel. Dir. : Serge Haroche (Prix Nobel 2012). Soutenu le 12 décembre 2009.
2005–2006	Master 2 Physique quantique, ENS Paris.
2002–2005	Diplôme de l'École Polytechnique (majeure Physique).
2000–2002	Classes préparatoires, admis à l'École Polytechnique. Lycée la Martinière Montplaisir, Lyon.

Financements (depuis 2014)

2023–2028	Ferbo (ANR PRCE) – partenaire – 150 k€
2021–2026	MecaFlux (ANR PRCE) – coordinateur principal – 640 k€
2019–2022	HyQuTech (Émergences Sorbonne Université) – coordinateur principal – 90 k€
2018–2022	CryoParis (SIRTEQ mi-lourd) – coordinateur principal – 280 k€ (cryostat à dilution)
2016–2020	OMT (H2020 ITN, Marie Curie) – partenaire (avec A. Heidmann)
2014–2019	QuNaT (ANR JCJC, ANR-14-CE26-0002) – coordinateur – 300 k€

Encadrement

Doctorants	M. Roussel (2026–, CIFRE LPENS/Alice & Bob, co-encadré avec Z. Leghtas) ; L. Godet (2026–, co-encadré avec T. Jacqmin) ; P. Manset (2023–) ; R. Rousseau (2022–25, CIFRE LKB/Alice & Bob) ; K. Gerashchenko (2021–25) ; H. Patange (2021–25) ; L. Najera-Santos (2020–24) ; É. Ivanov (2018–21) ; T. Capelle (2016–20) ; L. Neuhaus (2011–16, co-dév. PyRPL, dir. P.-F. Cohadon) ; T. Karassouloff (2012–16, dir. A. Heidmann) ; K. Makles (2011–15, dir. T. Briant).
Postdocs	L. Balembois (2023–25) ; Xu Chen (2015–17).
Stagiaires M2	Y. Li (2019) ; L. Bresque (2019) ; P. Manset (avant thèse) ; T. Capelle (M1, 2015) ; T. Karassouloff (2012).
Stagiaire L3	P. Rodriguez (2021).

Enseignement

2017–2024	M2 ICFP (ENS Paris) : *Quantum Machines* (un semestre/an ; co-enseignement P.-F. Cohadon puis P. Bertet).
2016–2017	M2 ICFP (ENS Paris) : *Quantum Optomechanics and Nanomechanics* (co-enseignement P.-F. Cohadon).
2024	*From gravitational wave detection to quantum optomechanics*, MesoSchool, Cargèse.
2022	*The New Mechanics*, École de Physique des Houches.
02/2013	Cours de spectroscopie (2 semaines), Université des Sciences et Techniques de Masuku (USTM), Gabon – programme PAI-DRH (CNRS).
2016–2020	Projets d'initiation à la programmation Python, Master 2 LUMI, Sorbonne Université (co-ens. T. Jacqmin, P. Cladé).
2019–2020	MOOC d'optomécanique, réseau européen OMT (en ligne).
2017–2018	SPOC de physique quantique (L3), Sorbonne Université (co-ens. T. Jacqmin, E. Lamour, Q. Glorieux).

2015–2021	Examinateur, épreuve de TP de physique, concours d'admission à Centrale-Supélec (filière MP). Développement de nouvelles expériences ; évaluation des candidats.
2010–2012	TD d'optique quantique, EPFL (Lausanne).
2007–2010	TD de physique quantique et statistique, ENSTA.
2006–2009	TD de thermodynamique et mécanique ; TP optique et électricité, ESIEE / Paris 6.

Valorisation

Brevet (inv.)	EP3899814B1 (2019) – *Single microwave photon detector* – CNRS/Sorbonne/ENS/Collège de France. Inv. : Lescanne, **S. Deléglise**, Flurin, Leghtas.
Brevet (CIFRE)	WO/2025/186341 (2025) – Alice & Bob – thèse CIFRE R. Rousseau, dir. S. Deléglise ; **S. Deléglise** non-inventeur.
Logiciel	PyRPL (Python Red Pitaya Lockbox) – contrôle FPGA open source pour l'optique quantique. *Rev. Sci. Instrum.* **95**, 033003 (2024). (17 cit.)

Articles (hors grandes collaborations)

$h = 17$; 32 articles publiés et preprints — grande collaboration Virgo/LIGO en fin de document

Liste complète. Les actes de conférences sont signalés [actes].

2026	Patange, Gerashchenko, Rousseau, Manset, Balembois, Capelle, **S. Deléglise**, Jacqmin, *Cryogenic rf-to-microwave transducer based on a dc-biased electromechanical system*, *Comptes Rendus. Physique* **27**, 49–63.
2025	Gerashchenko, Rousseau, Balembois, Patange, Manset, Smith, Leghtas, Flurin, Jacqmin, **S. Deléglise**, *Probing the quantum motion of a macroscopic mechanical oscillator with a radio-frequency superconducting qubit* ↗, arXiv :2505.21481, soumis à *Nature*. (3 cit.)
2025	Rousseau *et al.* (dont **S. Deléglise**), *Enhancing dissipative cat qubit protection by squeezing* ↗, arXiv :2502.07892, accepté (*Nat. Commun.*). (8 cit.)
2025	Manset, Palomo, Schmitt, Gerashchenko, Rousseau, Patange, Abgrall, Flurin, **S. Deléglise**, Jacqmin, Balembois, *Hyperinductance based on stacked Josephson junctions*, arXiv (2025). (5 cit.)
2024	Najera-Santos, Rousseau, …, **S. Deléglise**, *High-sensitivity AC-charge detection with a MHz-frequency fluxonium qubit* ↗, *Phys. Rev. X* **14**, 011007.
2024	Neuhaus, Croquette, Metzdorff, Chua, Jacquet, Journeaux, Heidmann, Briant, Jacqmin, Cohadon, **S. Deléglise**, *Python Red Pitaya Lockbox (PyRPL) : An open source software package for digital feedback control in quantum optics experiments*, *Rev. Sci. Instrum.* **95**, 033003. (17 cit.)
2023	Neuhaus, Croquette, Metzdorff, Chua, Jacquet, Journeaux, Heidmann, Briant, Jacqmin, Cohadon, **S. Deléglise**, *FPGA-based feedback control of quantum optics experiments with the open source software package PyRPL* ↗, arXiv :2310.00086. (6 cit.)

2021 Neuhaus *et al.* (dont **S. Deléglise**), *Laser cooling of a Planck mass object close to the quantum ground state* ↗, arXiv :2104.11648. (4 cit.)

2020 Lescanne, **S. Deléglise**, Albertinale, Réglade, Capelle, Ivanov, Jacqmin, Leghtas, Flurin, *Irreversible Qubit-Photon Coupling for the Detection of Itinerant Microwave Photons* ↗, *Phys. Rev. X* **10**, 021038. (139 cit.)

2020 Capelle, Flurin, Ivanov, Palomo, Rosticher, Chua, Briant, Cohadon, Heidmann, Jacqmin, **S. Deléglise**, *Probing a Two-Level System Bath via the Frequency Shift of an Off-Resonantly Driven Cavity* ↗, *Phys. Rev. Applied* **13**, 034022. (34 cit.)

2020 Ivanov, Capelle, Rosticher, Palomo, Briant, Cohadon, Heidmann, Jacqmin, **S. Deléglise**, *Edge mode engineering for optimal ultracoherent silicon nitride membranes*, *Appl. Phys. Lett.* **116**, 021102. (8 cit.)

2019 Lescanne, **S. Deléglise**, Albertinale, Réglade, Capelle, Ivanov, Jacqmin, Leghtas, Flurin, *Detecting itinerant microwave photons with engineered non-linear dissipation* ↗, arXiv :1902.05102. (14 cit.)

2018 Forni, Sarlette, Capelle, Flurin, **S. Deléglise**, Rouchon, *Adiabatic elimination for multi-partite open quantum systems with non-trivial zero-order dynamics*, *Proc. IEEE CDC*. (17 cit.)

2018 Bon, Neuhaus, **S. Deléglise**, Briant, Abbé, Cohadon, Galliou, *Cryogenic optomechanic cavity in low mechanical loss material*, *J. Appl. Phys.* **124**, 073104. (12 cit.)

2018 Bon, Rosenziveig, Abbé, Rocher, Galliou, Neuhaus, **S. Deléglise**, Briant, Cohadon, *Optomechanical coupling in a quartz crystal resonator for cryogenic clocks*, *Proc. IEEE IFCS*. (1 cit.)

2017 Chen, Chardin, Makles, Caër, Chua, Braive, Philip, Briant, Cohadon, Heidmann, Jacqmin, **S. Deléglise**, *High-finesse Fabry–Perot cavities with bidimensional Si_3N_4 photonic-crystal slabs* ↗, *Light : Science & Applications* **6**, e16190. (105 cit.)

2016 Garcia-Sanchez, **S. Deléglise**, Thomas, Atkinson, Lagoin, Perrin, *Acoustic confinement in superlattice cavities* ↗, *Phys. Rev. A* **94**, 033813. (18 cit.)

2016 Galliou *et al.* (dont **S. Deléglise**), *A new method of probing mechanical losses of coatings at cryogenic temperatures*, *Rev. Sci. Instrum.* **87**, 123906. (6 cit.)

2015 Makles *et al.* (dont **S. Deléglise**), *2D photonic-crystal optomechanical nanoresonator* ↗, *Optics Letters* **40**, 375. (37 cit.)

2014 Kuhn *et al.* (dont **S. Deléglise**), *Free-space cavity optomechanics in a cryogenic environment*, *Appl. Phys. Lett.* **104**, 044102. (25 cit.)

2012 Verhagen, **S. Deléglise**, Weis, Schliesser, Kippenberg, *Quantum-coherent coupling of a mechanical oscillator to an optical cavity mode* ↗, *Nature* **482**, 63–67. (1135 cit.)

2010 Weis, Rivière, **S. Deléglise**, Gavartin, Arcizet, Schliesser, Kippenberg, *Optomechanically induced transparency* ↗, *Science* **330**, 1520–1523. (1906 cit.)

2010 Rivière, **S. Deléglise**, Weis, Gavartin, Arcizet, Schliesser, Kippenberg, *Optomechanical sideband cooling of a micromechanical oscillator close to the quantum ground state* ↗, *Phys. Rev. A* **83**, 063835. (228 cit.)

2010 Gorodetsky, Schliesser, Anetsberger, **S. Deléglise**, Kippenberg, *Determination of the vacuum optomechanical coupling rate using frequency noise calibration* ↗, *Optics Express* **18**, 23236. (244 cit.)

2008 **S. Deléglise**, Dotsenko, Sayrin, Bernu, Brune, Raimond, Haroche, *Reconstruction of non-classical cavity field states with snapshots of their decoherence* ↗, *Nature* **455**, 510–514. (829 cit.)

2008 Bernu, **S. Deléglise**, Sayrin, Kuhr, Dotsenko, Brune, Raimond, Haroche, *Freezing a Coherent Field Growth in a Cavity by Quantum Zeno Effect* ↗, *Phys. Rev. Lett.* **101**, 180402. (164 cit.)

2008 Brune, Bernu, Guerlin, **S. Deléglise**, Sayrin, Gleyzes, Kuhr, Dotsenko, Raimond, Haroche, *Process tomography of field damping and measurement of Fock state lifetimes by quantum non-demolition photon counting in a cavity* ↗, *Phys. Rev. Lett.* **101**, 240402.

2007 Gleyzes, Kuhr, Guerlin, Bernu, **S. Deléglise**, Busk Hoff, Brune, Raimond, Haroche, *Quantum jumps of light recording the birth and death of a photon in a cavity* ↗, *Nature* **446**, 297–300. (866 cit.)

2007 Guerlin, Bernu, **S. Deléglise**, Sayrin, Gleyzes, Kuhr, Brune, Raimond, Haroche, *Progressive field-state collapse and quantum non-demolition photon counting* ↗, *Nature* **448**, 889–893. (715 cit.)

2007 Kuhr, Gleyzes, Guerlin, Bernu, Busk Hoff, **S. Deléglise**, Osnaghi, Brune, Raimond, Haroche, Jacques, Bosland, Visentin, *Ultrahigh finesse Fabry-Perot superconducting resonator*, *Appl. Phys. Lett.* **90**, 164101. (310 cit.)

Exposés invités en conférences (depuis 2011)

06/2027 *À venir* — FNS 2027, Frontiers of Nanomechanical Systems, INL Braga, Portugal (inv. S. Hanay).

04/2027 *À venir* — DK ALM Physics Colloquium, Universität Innsbruck, Autriche.

11/2026 *Probing the quantum motion of a MHz mechanical resonator with a resonant rf-fluxonium*, Séminaire LOMA, Bordeaux (inv. F. Pistolesi).

03/2026 *idem*, Séminaire ICFO, Castelldefels, Barcelone (inv. A. Bachtold).

03/2026 *Probing the quantum motion of a macroscopic mechanical resonator with a resonant rf-fluxonium*, Hybrid Quantum System Workshop, Ascona (inv. Y. Chu).

01/2026 *idem*, Séminaire LIP, ENS Lyon (inv. A. Bienfait).

10/2025 *idem*, Symposium 40 ans de l'optique quantique, Paris (inv. P.-F. Cohadon).

04/2025 *idem*, Séminaire CEA/SPEC, Saclay (inv. E. Flurin).

12/2024 *Interfacing a macroscopic mechanical resonator with a MHz-Fluxonium qubit*, GDR Physique mésoscopique, Aussois (inv. H. Pothier).

07/2024 *MHz-fluxonium qubit for quantum control of mechanical systems*, Congrès SFO, Rouen (inv. C. Champenois).

03/2024 *idem*, Gordon Research Conf. on Mechanical Systems in the Quantum Regime, Ventura CA (inv. A. Safavi-Naeini).

12/2023 *Towards quantum control of an ultracoherent mechanical resonator with a RF-fluxonium qubit*, Séminaire NEQ, Institut Néel, Grenoble (inv. J. Viennot).

02/2022 *A Mechanics-based Microwave to Optical Quantum Link*, Kick-off CNRS/UArizona IRC, Tucson (inv. R. Ferrière).

01/2019 *Electromechanics – towards heralded non-classical motional states*, OMT/HOT Annual Meeting, Gstaad.

10/2018 *Optomechanical systems in the quantum regime*, SMYLE Kick-off Meeting, Neuchâtel.

01/2018 *Tutorial on Optomechanical Resonators*, OMT/HOT Annual Meeting, Gstaad.

07/2017 *Quantum optomechanics – experiments and applications*, Congrès annuel SFP, Orsay.

03/2017 *Superconducting resonators in quantum engineering*, Workshop on Axion Physics, Rome.

02/2017 *Photonic crystal slabs for optomechanics*, SPIE Photonics West, San Francisco.

04/2016 *Quantum effects in optomechanical systems*, LIP2016, Xi'An, Chine.

09/2013 *Optomechanics with micro and nano-mirrors*, Frontiers of Nanomechanics, ICTP, Trieste.

07/2013 *Refroidissement optique de systèmes mécaniques*, Congrès national SFP, Marseille.

08/2012 *Cavity optomechanics – towards quantum control of macroscopic devices*, JMC13, Montpellier.

11/2012 *Électrodynamique et optomécanique quantique en cavité*, Séance de l'Académie des sciences, Paris.

07/2011 *Cavity optomechanics – towards quantum control of micromechanical motion with light*, NLO Conference, Kaua'i, Hawaii.

01/2011 *Optomécanique avec des microcavités de haute finesse*, Séminaire du cours de S. Haroche, Collège de France, Paris.

Séminaires

02–05/2009 *Preparation and reconstruction of non-classical field states in Cavity QED*, séminaires invités : Martinis (UCSB), Vuckovic (Stanford), Jun Ye (Boulder), Wineland (Boulder), Vuletic (MIT), Schoelkopf (Yale), MPQ (Garching), Hänsch (LMU Munich), ETHZ (Zürich), SPEC (Saclay), GAP (Genève). (11 séminaires)

11/2012 *Électrodynamique et optomécanique quantique en cavité – vers le contrôle quantique d'objets macroscopiques à l'aide de la lumière*, Séminaire FEMTO-ST, Besançon (inv. S. Galliou).

03/2008 *Preparation and reconstruction of Schrödinger cat states in cavity QED*, Young Atom Opticians Conference, Florence.

11/2008 *Quantum Zeno effect in a cavity QED experiment* [poster], Symposium on Physics of Quantum Technology, Nara, Japon.

09/2007 *Light under repeated observation — Fock states and quantum jumps*, Young European Physicist Meeting, Frauenchiemsee, Allemagne.

Communications en congrès (sélection)

03/2023 *Sideband cooling of a Radio-Frequency superconducting qubit*, APS March Meeting, Las Vegas.

08/2021 *Hybrid superconducting-mechanical circuits*, 17es Journées de la Matière Condensée (JMC17) (distanciel).

04/2017 *Hybrid Nanomechanical Transducers*, Quantum Information and Measurement IV (QIM-IV), Paris.

06/2017 *Hybrid optomechanical systems as transducers for quantum information*, CLEO Europe, Munich.

07/2015 *Optomechanics and Quantum Measurements*, Shanghai Meeting on Nanophononics, Tongji University.

2013 *Towards observation of quantum optomechanical correlations*, CLEO Europe.

2011 *Cavity optomechanics — Cooling of a micromechanical oscillator into the quantum regime*, 16th International Conference on Optical MEMS and Nanophotonics.

2009 *Preparation and reconstruction of non-classical field states in a cavity QED experiment*, CLEO/QELS, USA.

Responsabilités institutionnelles

2024–2026 Membre du bureau, UFR de physique, Sorbonne Université.

2018–2024 Responsable des séminaires du LKB (Paris) : sélection des orateurs, premier contact, chairman, organisation des visites de laboratoire.

2017–2026 Membre du bureau ColoQ (SFO) ; sélection des contributions au congrès national.

2015–2017 Expert membre de l'OMNT (Observatoire des Micro et Nanotechnologies, CNRS/CEA) ; veille technologique, environ une réunion par mois.

2019 Membre du jury de recrutement MCF, Sorbonne Université (Section CNU 30, poste n°1616).

2017– Membre de la Commission de Spécialiste (Section 30), Université Pierre et Marie Curie.

2016– Correspondant International de l'INP (CNRS Institut de Physique).

Jury de thèse

11/2025 Jean-Paul van Soest, TU Delft (gr. G. Steele) – examinateur.

2022 Yannick Seis, Université de Copenhague (NBI) – *opponent*.

11/2019 Tom Delord-Carnaut, ENS Paris (gr. G. Hetet) – examinateur.

05/2019 Jan Bilek, DTU (gr. U. Andersen) – examinateur.

2015–16 Jiayi Qin, Université d'Australie Occidentale – rapporteur.

2015 Yiqiu Ma, Université d'Australie Occidentale – rapporteur.

Jury de stages

2021 L. Pallegoix, M2 ICFP (gr. E. Flurin & P. Bertet, CEA/Saclay).

2021 A. Vincenti, L3 FIP (gr. O. Arcizet & B. Pigeau, Institut Néel).

2021 E. Seraille, L3 FIP (gr. P. Cladé, LKB).

2016 T. Peronnin, M2 ICFP (gr. B. Huard, LPA/ENS Paris).

2015 V. Métillon, M2 (gr. I. Dotsenko, LKB Paris).

2014 D. Marković, M2 ICFP (gr. B. Huard, LPA/ENS Paris).

Referee de journaux

Nature, *Nature Photonics*, *Phys. Rev. Lett.*, *Phys. Rev. X*, *Phys. Rev. A*, *Phys. Rev. Applied*, *Optica*, *Optics Express*, *EPL*, *EPJD*.

Vulgarisation scientifique

2012–2021 Participation régulière à la Fête de la science au LKB (Laboratoire Kastler Brossel).

2017 *La Nuit des ondes gravitationnelles* – membre du comité d'organisation ; soirée festive simultanée dans 14 villes (12 France, 2 Italie), grand public, diffusion en direct depuis le Grand Rex (Paris) avec visite en direct de la salle de contrôle de Virgo (Cascina).

2017 **S. Deléglise** *et al.*, *Un dispositif optique dédié à l'information quantique*, *La Recherche* n°526–527, p. 52.

2012 **S. Deléglise**, *Le monde quantique à notre échelle*, *La Recherche* (numéro spécial), août 2012.

Prix et distinctions

2016 Gruber Cosmology Prize (collab. LIGO–Virgo).

2016 Breakthrough Prize in Fundamental Physics (collab. LIGO–Virgo).

2012 Prix Gustave Ribaud (Académie des sciences) – refroidissement optique de résonateurs micromécaniques.

Publications au sein de la grande collaboration (Virgo/LIGO) 148 publications

Samuel Deléglise est membre de la collaboration Virgo (2009–2025). Liste complète par ordre chronologique inverse.

2026 *Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of LIGO and Virgo.*

2025 *GWTC-4.0 : An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog.* (35 cit.)
GWTC-4.0 : Methods for Identifying and Characterizing Gravitational-wave Transients. (24 cit.)
GWTC-4.0 : Population Properties of Merging Compact Binaries. (11 cit.)

Optical characterization of the Advanced Virgo gravitational wave detector for the O4 observation run, *Applied optics.*
Search for continuous gravitational waves from known pulsars in the first part of the fourth LIGO-Virgo-KAGRA observing run. (19 cit.)

2024 *A search using GEO600 for gravitational waves coincident with fast radio bursts from SGR 1935+2154.* (12 cit.)
Observation of Gravitational Waves from the Coalescence of a 2.5-4.5 $M_\odot$ *Compact Object and a Neutron Star.*
Search for gravitational waves emitted from SN 2023ixf. (19 cit.)

Swift-BAT GUANO follow-up of gravitational-wave triggers in the third LIGO-Virgo-KAGRA observing run. (5 cit.)
Ultralight vector dark matter search using data from the KAGRA O3GK run. (28 cit.)

2023 *A Joint Fermi-GBM and Swift-BAT Analysis of Gravitational-Wave Candidates from the Third Gravitational-wave Observing Run.* (14 cit.)
Constraints on the cosmic expansion history from GWTC-3. (369 cit.)

Open data from the third observing run of LIGO, Virgo, KAGRA and GEO. (357 cit.)
Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo. (57 cit.)
Search for gravitational-lensing signatures in the full third observing run of the LIGO-Virgo network. (117 cit.)
The population of merging compact binaries inferred using gravitational waves through GWTC-3, *Physical Review X.*

2022 *All-sky search for continuous gravitational waves from isolated neutron stars using Advanced LIGO and Advanced Virgo O3 data.* (179 cit.)
First joint observation by the underground gravitational-wave detector, KAGRA, with GEO600.
Model-based cross-correlation search for gravitational waves from the low-mass X-ray binary Scorpius X-1 in LIGO O3 data. (40 cit.)
Search for Gravitational Waves Associated with Fast Radio Bursts Detected by CHIME/FRB During the LIGO–Virgo Observing Run O3a. (46 cit.)
Search for continuous gravitational wave emission from the Milky Way center in O3 LIGO–Virgo data. (93 cit.)
Search for gravitational waves from Scorpius X-1 with a hidden Markov model in O3 LIGO data. (46 cit.)

Search for gravitational-wave transients associated with magnetar bursts in Advanced LIGO and Advanced Virgo data from the third observing run. (46 cit.)
Search for subsolar-mass black hole binaries in the second part of Advanced LIGO's and Advanced Virgo's third observing run. (125 cit.)
The Virgo O3 run and the impact of the environment. (34 cit.)

Virgo Detector Characterization and Data Quality during the O3 run. (20 cit.)

Virgo Detector Characterization and Data Quality : results from the O3 run. (73 cit.)
Virgo Detector Characterization and Data Quality : tools. (20 cit.)

2021 *All-sky Search for Continuous Gravitational Waves from Isolated Neutron Stars in the Early O3 LIGO Data.* (83 cit.)
All-sky search for gravitational wave emission from scalar boson clouds around spinning black holes in LIGO O3 data. (116 cit.)
All-sky search for long-duration gravitational-wave bursts in the third Advanced LIGO and Advanced Virgo run.
All-sky search for short gravitational-wave bursts in the third Advanced LIGO and Advanced Virgo run. (154 cit.)
All-sky, all-frequency directional search for persistent gravitational-waves from Advanced LIGO's and Advanced Virgo's first three observing runs. (54 cit.)
Calibration of Advanced Virgo and reconstruction of detector strain h(t) during the Observing Run O3. (75 cit.)
Constraints from LIGO O3 data on gravitational-wave emission due to r-modes in the glitching pulsar PSR J0537-6910. (73 cit.)
Constraints on cosmic strings using data from the third Advanced LIGO-Virgo observing run. (291 cit.)
Constraints on dark photon dark matter using data from LIGO's and Virgo's third observing run. (122 cit.)
GWTC-2.1 : Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run. (835 cit.)
GWTC-3 : Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run. (3366 cit.)
Narrowband searches for continuous and long-duration transient gravitational waves from known pulsars in the LIGO-Virgo third observing run. (104 cit.)
Observation of gravitational waves from two neutron star-black hole coalescences. (1047 cit.)
Search for Gravitational Waves Associated with Gamma-Ray Bursts Detected by Fermi and Swift During the LIGO-Virgo Run O3b. (90 cit.)
Search for anisotropic gravitational-wave backgrounds using data from Advanced LIGO and Advanced Virgo's first three observing runs.
Search for continuous gravitational waves from 20 accreting millisecond X-ray pulsars in O3 LIGO data. (291 cit.)
Search for lensing signatures in the gravitational-wave observations from the first half of LIGO-Virgo's third observing run. (232 cit.)
Search for subsolar-mass binaries in the first half of Advanced LIGO and Virgo's third observing run.
Search of the Early O3 LIGO Data for Continuous Gravitational Waves from the Cassiopeia A and Vela Jr. Supernova Remnants. (50 cit.)

Searches for Gravitational Waves from Known Pulsars at Two Harmonics in the Second and Third LIGO-Virgo Observing Runs. (106 cit.)
Searches for continuous gravitational waves from young supernova remnants in the early third observing run of Advanced LIGO and Virgo. (85 cit.)
Tests of General Relativity with GWTC-3. (423 cit.)

Upper Limits on the Isotropic Gravitational-Wave Background from Advanced LIGO's and Advanced Virgo's Third Observing Run. (645 cit.)

2020 *A Joint Fermi-GBM and LIGO/Virgo Analysis of Compact Binary Mergers From the First and Second Gravitational-wave Observing Runs.* (29 cit.)
All-sky search in early O3 LIGO data for continuous gravitational-wave signals from unknown neutron stars in binary systems. (88 cit.)
Diving below the spin-down limit : Constraints on gravitational waves from the energetic young pulsar PSR J0537-6910. (190 cit.)
GW190412 : Observation of a Binary-Black-Hole Coalescence with Asymmetric Masses. (830 cit.)
GW190425 : Observation of a Compact Binary Coalescence with Total Mass $\sim 3.4 M_{\odot}$.
GW190521 : A Binary Black Hole Merger with a Total Mass of 150 $M_{\odot}$. (1576 cit.)
GW190814 : Gravitational Waves from the Coalescence of a 23 $M_{\odot}$ *Black Hole with a 2.6* $M_{\odot}$ *Compact Object.*
GWTC-2 : Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run. (3025 cit.)
Gravitational-wave constraints on the equatorial ellipticity of millisecond pulsars. (110 cit.)
Population Properties of Compact Objects from the Second LIGO-Virgo Gravitational-Wave Transient Catalog. (1150 cit.)
Properties and astrophysical implications of the 150 Msun binary black hole merger GW190521. (745 cit.)
Quantum backaction on Kg-scale mirrors : observation of radiation pressure noise in the advanced Virgo detector, *Physical Review Letters.* (77 cit.)
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